Real-Time Traffic Analysis Report

Key Finding

The data reveals that Cairo’s road network experiences severe congestion even during weekend hours, with an average congestion ratio of 89.4% and an average speed of only 30.06 km/h during the Saturday 4:00 PM peak period. This is a strong indicator that the city’s infrastructure is under significant pressure regardless of whether it is a workday or a weekend.

The maximum recorded speed was 78 km/h, observed on major highways such as the 26th of July Corridor and Ring Road, while the minimum speed dropped to just 9 km/h on inner-city streets like El-Gaish Street and El-Gomhoreya Street, reflecting extreme gridlock in central districts.

Traffic Hotspots

The most congested areas are concentrated in Downtown Cairo, Old Cairo, and Giza, where narrow road networks combined with high population density create bottlenecks that are difficult to resolve without structural intervention. In contrast, highways and ring roads perform significantly better, demonstrating that traffic naturally seeks wider, less interrupted routes when available.

Recommendations

1. Smart Signal Management

Deploying AI-powered adaptive traffic signals in high-congestion zones such as Downtown and Dokki would dynamically adjust green light durations based on real-time traffic density, reducing unnecessary wait times and improving flow.

2. Traffic Redistribution

Encouraging drivers to use alternative corridors such as the 26th of July Corridor, Salah Salem Road, and the Ring Road during peak hours would relieve pressure on central streets. This can be achieved through real-time navigation app integration and clear road signage.

3. Public Transportation Enhancement

Increasing the frequency of buses and metro services along the Shobra–Downtown–Giza axis during peak hours would provide a reliable alternative to private vehicles, gradually reducing road load.

4. Flexible Scheduling Policies

Given that congestion reaches critical levels even on weekends, businesses and commercial centers located in Downtown and Giza should be encouraged to stagger operating hours, spreading traffic across a wider time window rather than concentrating it around 4:00 PM.

5. Long-Term Infrastructure Planning

The fact that weekend congestion mirrors workday patterns suggests a deeper structural issue. A comprehensive urban mobility study is recommended to identify opportunities for road expansion, pedestrian zone creation, and smarter land-use planning in central Cairo.

Conclusion

Cairo's traffic congestion is not limited to working hours — it is a persistent urban challenge that requires both immediate operational measures and long-term strategic planning. The data clearly shows that inner-city streets are operating beyond their capacity, while highways remain underutilized relative to their potential. Bridging this gap through smart technology, better public transport, and informed urban policy is essential for improving mobility across Greater Cairo.